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LLM Strategies

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Best Practices

No.	Practice	Explanation	Example
1	Explicitly ask for best practices	Explicitly request recommended approaches or best practices.	“I want to (…) – What approach would you recommend?”
2	Assign a role to ChatGPT	Define the role or persona from whose perspective the response should be generated.	“You are an experienced Python developer applying current industry best practices …”
3	Describe the goal and ask for an implementation strategy	Clearly describe the target outcome. For complex tasks, split the goal into multiple requests.	An existing Excel VBA tariff calculator should be migrated to Python.
4	Provide constraints and context	Supplying contextual information and constraints improves the quality of the solution.	Specify required modules, technical edge cases such as non-standard delimiters, etc.
5	Instruct the model to ask clarification questions	Explicitly ask the model to clarify unknown parameters or functions.	“If parameters or functions are unclear or undefined, ask clarification questions.

Best Practices

No.	Practice	Explanation	Example
6	Explicitly address known prompt pitfalls	Anticipate known pitfalls and define the desired behavior directly in the prompt.	Explicitly mention non-standard delimiters in the prompt.
7	Specify the expected output format precisely	Providing examples (e.g., target data structures) helps align generated code with the intended structure. This can be done via code, detailed descriptions, or screenshots. **Risk**: Over-specification may prevent alternative or more efficient formats	Create a list and output it as XML.
8	Apply corrections within the chat	Corrections made outside the chat are not reflected in the model context and must be manually replicated. Balance this against LLM request limits.	
9	Use separate chats for independent subtasks (“fresh context”)	If results degrade, summarize previous results and start a new chat with a clean context.	

Two Strategies – Overview

Crafted Approach

Goal: Rapid prototyping – with as few prompts as possible.

Approach:

- Since the reasoning model cannot process Excel files, the individual components of the Excel tariff calculator (tables, formulas, VBA code) are handled separately.
- From this, the equivalent Python tariff calculator is incrementally generated through appropriate prompting.

Industrial Approach

Goal: 100% script-driven migration – no manual intermediate steps.

Approach:

- Python programs are generated that can automatically extract the relevant information from the Excel tariff calculator and transform it into a standardized text or CSV format.
- Based on this format, the equivalent Python tariff calculator is generated. The generation is performed via a sequence of prompts that can be executed by a fully automated prompt pipeline.

Craft-Based Approach – Workflow

Dependencies: The “Calculation” worksheet depends on VBA code, and the VBA code depends on mortality tables.

This results in the following workflow:

1. Transfer mortality tables into a machine-readable format
2. Translate VBA code into functional Python code
3. Port the worksheet logic and calculations to Python
4. For quality assurance: build a Python validation script to compare results

Industrial Approach – Workflow

1. **Automated extraction from Excel (Excel dump)**

   Use Python scripts to extract all relevant information from Excel. These scripts are

   - generated once using ChatGPT
   - produce standardized outputs: in CSV format (for cell contents and ranges) and in TXT format (for VBA code)

2. **Translate core functions (VBA) to Python**
   - Input: VBA modules as TXT files (from step 1)
   - Output: equivalent Python modules and corresponding tests

3. **Translate output functions, such as premiums and projection values, to Python**
   - Input: CSV files (from step 1) and Python core modules (from step 2)
   - Output: equivalent Python modules and corresponding tests

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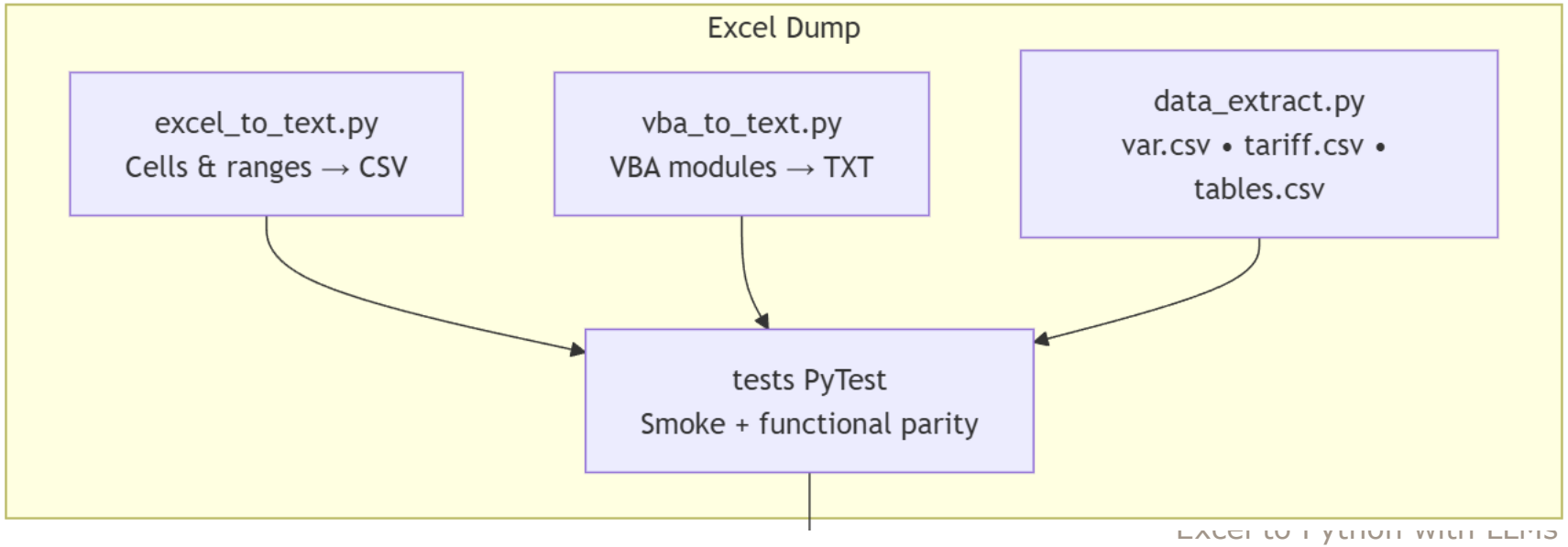

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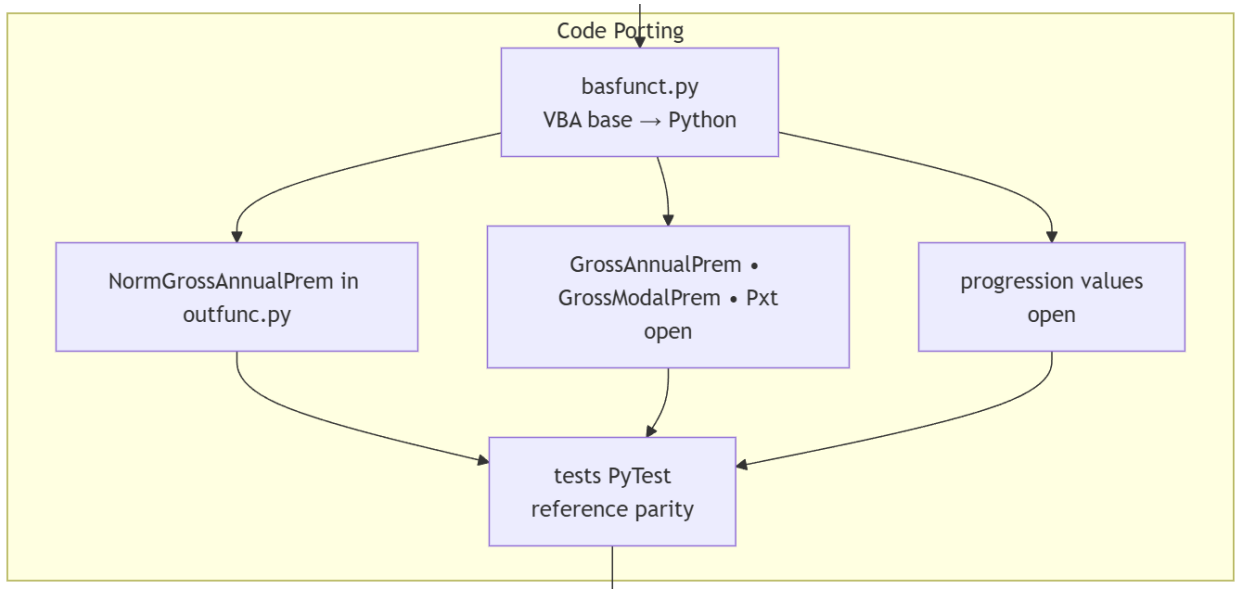
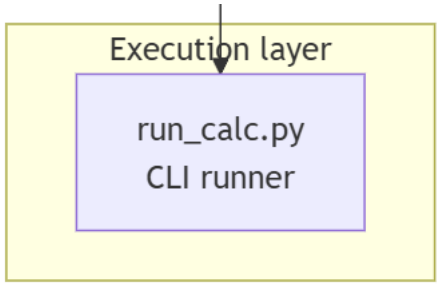
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